

Mixed Integer Programming

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Lecture 2

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1 Binary variables

- Can take the values 0 and 1 only
- For most real-world applications, binary variables are necessary in order to obtain a realistic model

Example: an investment problem

- A company can choose between the following investment projects: 1, ..., 8
- In total, the company can invest up to 100M
- The company wants to maximize total NPV
- Each project is of the type “all in or nothing”, i.e., partial investment in the project is not possible
- This is a variation of the knapsack problem

1.1 Problem with only binary variables

A manufacturer of electronic components has limited production capacity and needs to prioritize which customer orders to satisfy during periods of high demand.

The company wants to select orders so as to maximize total profit. Each order must either be satisfied completely or rejected.

On a given day, the manufacturer must choose between the following orders:

	Revenue	Cost	Hours per order
Customer 1	18	8	3
Customer 2	32	9	7
Customer 3	25	9	5
Customer 4	24	10	4
Customer 5	26	13	3

“Hours per unit” is the number of production hours spent for the order.

There are 16 hours available in the production facility

Parameters:

- r_i : revenue per order
- c_i : cost per order
- a_i : hours per order
- b : capacity

Variables

$$X_i = \begin{cases} 0 & \text{if we reject order } i \\ 1 & \text{otherwise} \end{cases}$$

Model

$$\begin{aligned} & \max \left(\sum_i (r_i - c_i) X_i \right) \\ & \text{s.t.} \\ & \sum_i a_i X_i \leq b \\ & X_i \in \{0, 1\} \end{aligned}$$

1.2 Model with both continuous and binary variables

Mixed-Integer Problem: MIP

A company needs to decide production quantities of five product types so to maximize total profit.

For each product type, there is a market demand that defines the maximum possible quantity sold

Now we have added a fixed cost for each product type. The cost is avoided if the product type is not produced

The five product types share a joint production capacity that limits the total production:

	Sales price per unit	Variable cost per unit	Hours per unit produced	Max number of units sold	Fixed cost
Type 1	18	8	0.30	2500	3700
Type 2	32	9	0.70	3000	1900
Type 3	25	9	0.50	4000	1000
Type 4	24	10	0.40	2000	2200
Type 5	26	13	0.30	1500	2600

There are 2000 hours available for production

Parameters

- p_i : price per unit
- c_i : cost per unit
- h_i : hours per unit
- m_i : max amount per unit
- b : capacity
- $\mathcal{F}_i$: fixed cost

Variables

- X_i : quantity

$$\lambda_i = \begin{cases} 0 & \text{if type is not produced} \\ 1 & \text{otherwise} \end{cases}$$

Model

$$\begin{aligned} \max \quad & \left(\sum_i ((p_i - c_i)X_i + \mathcal{F}_i)\lambda_i \right) \\ \text{s.t.} \quad & 0 \leq X_i \leq m_i \quad \forall i \\ & \sum_i h_i X_i \leq b \quad \forall i \\ & X_i + \lambda_i \end{aligned}$$